Tutorial-2

—pn junction—

1. A silicon pn junction employs $N_A = 2\times10^{16}$ cm^{-3} and $N_D = 4\times10^{16}$ cm^{-3} . Determine the built-in potential at room temperature (T = 300 K).
2. Barrier potential V_0 is a weak function of the doping levels. How much does it change if N_A or N_D is increased by one order of magnitude?
3. In a pn junction, find the majority and minority charge carriers and also find the W, x_n and x_p if $N_A = 10^{18}$ cm^{-3} and $N_D = 10^{16}$ cm^{-3} at T=300 K. (ϵ_s=1.04 $\times10^{-12}$ F/cm)
4. For Q1 determine I_S for the junction if A =100 μm^2 , L_n= 20 μm, and L_p=30 μm.

$$I_S = A q n_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right)$$

$D_n = 35\ cm^2/s$ $\quad$ $D_p = 12\ cm^2/s$

(1) $$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \qquad V_T \simeq 26mV$$

$$= 0.026 \ln\left(\frac{2\times10^{16} \times 4\times10^{16}}{(1.5\times10^{10})^2}\right)$$

$$= 751 mV$$

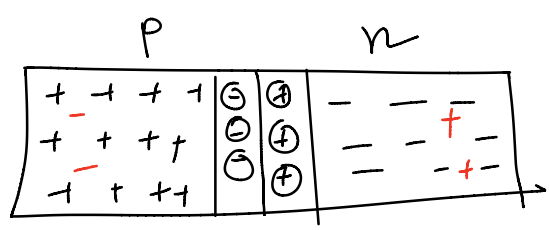
(2) $$\Delta V_0 = V_T \ln\left(\frac{10 N_A N_D}{n_i^2}\right) - V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$= V_T \ln\left(\frac{10 \cancel{N_A N_D}}{\cancel{n_i^2}} \times \frac{\cancel{n_i^2}}{\cancel{N_A N_D}}\right)$$

$$= 0.026 \ln 10 \quad \text{at } T = 300\text{K}$$

$$\approx \underline{\underline{60\,\text{mV}}}$$

(3)



$$p_p \simeq N_A = 10^{18}\,\text{cm}^{-3}$$

$$n_p \simeq \frac{n_i^2}{N_A} = \frac{(1.5\times10^{10})^2}{10^{18}} = 2.25\times10^{2}\,\text{cm}^{-3}$$

$$n_n \simeq N_D = 10^{16}\,\text{cm}^{-3}$$

$$p_n \simeq \frac{n_i^2}{N_D} = \frac{(1.5\times10^{10})^2}{10^{16}} = 2.25\times10^{4}\,\text{cm}^{-3}$$

$$W = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right) V_0}$$

$$= \sqrt{\frac{2 \times 1.04 \times 10^{-12} F/cm}{1.6 \times 10^{-19} C}\left(\frac{1}{10^{18}} + \frac{1}{10^{16}}\right) \times 0.817 V}$$

$$= 3.27 \times 10^{-7} m$$

$$= 0.327 \mu m$$

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

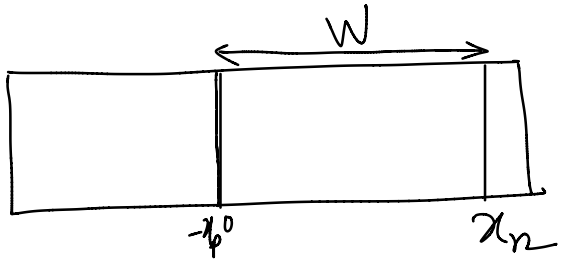
$$= 0.026 \ln\left(\frac{10^{18} \times 10^{16}}{(2.25 \times 10^{10})^2}\right)$$

$$= 0.817 V$$

$$\therefore x_n = \frac{N_A}{N_A + N_D} W = 0.324 \mu m$$

$$\therefore x_p = \frac{N_D}{N_A + N_D} W = 0.003 \mu m$$

$$x_n \approx W$$



(4) $$I_s = Aq n_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right)$$

$$A = 100\mu m^2 = 10^2 \times (10^{-4} cm)^2$$
$$= 10^{-6} cm^2$$

$$L_n = 20 \times 10^{-4} cm \qquad L_p = 30 \times 10^{-4} cm$$

$$\therefore I_s = 10^{-6} \times 1.6 \times 10^{-19} \times (1.5 \times 10^{10})^2 \left(\frac{35}{2 \times 10^{16} \times 20 \times 10^{-4}} + \frac{12}{4 \times 10^{16} \times 30 \times 10^{-4}} \right)$$

$$= \frac{1.6 \times 1.5^2 \times 10^{-5}}{10^{-12}} \left(\frac{\cancel{35}^{7}}{\cancel{40}_{8}} + \frac{\cancel{12}}{\cancel{120}_{10}} \right)$$

$$= 3.51 \times 10^{-17} A$$